Abeer Miglani

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Summary

Second-year Electrical and Computer Engineering student seeking a Software Engineering internship. Builds systems software in C++, C, and Python, with hands-on work on network protocols, graph algorithms, and Rust performance extensions. Also open to Embedded Systems and Junior Developer roles.

Education

Shiv Nadar Institution of Eminence Dadri, UP
Bachelor of Technology in Electrical and Computer Engineering *August 2025 – May 2029*

Experience

Summer Trainee GitHub | Jun. 2023
LOHUM Cleantech Pvt. Ltd. *Kasna, UP*

- Automated price collection for **9** lithium, nickel, and battery material products with Python and Selenium.
- Handled dynamically loaded pages with XPath selectors and explicit waits, and cleaned the data with Pandas.
- Generated an HTML dashboard from the scraped data to view and share prices in one place.

Projects

Ripple, Cascading Failure Simulator - *Rust, Python, React* GitHub | Sep. 2026

- Built solo at Manipal Hackathon 2026 to simulate failure spread across a synthetic **120-asset** city network.
- Made all-pairs shortest path **25x** faster, and **33x** on 2,000-node graphs, with a parallel Rust extension on PyO3.
- Cut worst-case recommendation latency from **1.5 s to 190 ms**; each request runs up to 50 all-pairs passes.
- Built a Motter-Lai cascade engine and a recommender whose top verified fix prevents up to **91%** of failures.
- Added **300** pytest tests and GitHub Actions CI that checks the Rust and Python results match.

Redis-Style Server - *C++, POSIX Sockets, TCP* GitHub | Jul. 2026 – Present

- Building a Redis-style in-memory server from first principles in C++ on raw POSIX sockets.
- Designed a length-prefixed binary protocol that survives fragmented TCP reads, tested with **1,000** split requests.
- Benchmarked **25K requests per second** at **38** μ**s** median and **82** μ**s** p99 latency over 100K loopback round trips.

Technical Skills

Languages: C++, C, Python, SQL, TypeScript, Rust, HTML, CSS
Frameworks and Libraries: FastAPI, Flask, React, Celery, NetworkX, PyO3, Rayon, Pandas, NumPy, Selenium
Databases: PostgreSQL, PostGIS, Neo4j, Redis, MySQL
Developer Tools: Git, Docker, Docker Compose, GitHub Actions, pytest, VS Code
Core Areas: Systems Programming, Computer Networks, Data Structures, Algorithms

Problem Solving

- Solved 140+ data structures and algorithms problems on LeetCode, Codeforces, CodeChef, and HackerRank.
- Solved 3 of 4 problems in an intra-college competitive programming contest.